

SHIVAM ARORA

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EDUCATION

University of Michigan

Ann Arbor, MI

M.S. in Computer Science & Engineering GPA: 3.83/4.00

Aug 2025 – Expected Dec 2026

Relevant Coursework: Applied GPU Programming, Advanced Scalable Systems for Agentic AI, Reinforcement Learning, Advanced Computer Vision, Probability & Random Processes, Computational Complexity (A+), Human-AI Interaction (A+), Learning Enabled Systems (Fall 2026)

Thapar Institute of Engineering and Technology

Punjab, India

B.E. in Computer Science; GPA: 3.98/4.00 (9.77/10.00)

Aug 2017 – Jun 2021

TECHNICAL SKILLS

Languages & Programming: C; C++; Python (NumPy, Pandas, OpenCV, FastAPI, NetworkX); Java (Spring, DropWizard); JavaScript; CUDA

ML & AI: PyTorch; TensorFlow; TorchRL; vLLM; LLMs (Qwen-2.5, Llama 3.1); Computer Vision (ViT, VLMs, CLIP, DINO); 3D/4D Human Mesh Reconstruction; Reinforcement Learning (MARL, MAPPO/IPPO, PPO, DQN); Agentic Workflows; Prompt Engineering; Semantic Normalization; Explainable AI (MAIA: Multimodal Automated Interpretability Agent); Knowledge Graphs (Neo4j)

Systems & Data: GPU Architecture; HPC; Concurrent Systems; DAG Scheduling; AWS (Glue, Lambda, S3); Apache Spark; Snowflake; HDFS; Terraform

Frameworks & Tools: React; MLflow; W&B; Git; JUnit; Reladomo; Legend (GS); Sybase IQ; Arduino

WORK EXPERIENCE

Goldman Sachs (Jan 2021 – July 2025)

Bengaluru, India

Associate - Technology

Jan 2024 – July 2025

- Led architecture of a **Spark, AWS Glue, and Snowflake-based** insurance calculator, **processing 6M+ deposit accounts daily with a 20-minute SLA**, enabling accurate pre-9AM ET calculations for timely deposit record keeping.
- Architected one of the **firm's first cloud-native AWS data pipelines** to process US deposit information, including **customer PII**; **improved data precision by 99%**, standardized data across divisions, and ensured **FDIC Part 370 compliance**.

Analyst - Technology

Jul 2021 – Dec 2023

- **Automated and streamlined** end-to-end German Deposit Protection and European regulatory reporting systems using **Java, SQL, REST APIs, and Sybase IQ**.
- **Improved the EdB regulatory rating from E (Fail) to B** and **reduced quarterly reconciliation effort by 60%**; received the **GS Controllers India Award (2022)** for exceptional performance in architecting the solution.

Intern - Technology

Jan 2021 – Jun 2021

- Developed a full-stack reconciliation platform using **React and Java Spring**, automating reconciliations between the firm's General Ledger and Regulatory Reporting data; served **500+ Controllers** and saved **800+ person-hours each quarter**.
- **Transitioned 300+ legacy screens to the modern GS Web stack** with a comprehensive **JUnit testing framework**; collaborated in an **Agile environment** and earned a **return offer**.

SELECTED RESEARCH & TECHNICAL PROJECTS

Dependency-Aware Execution for LLM Knowledge Graphs Advisor: [Dr. Mosharaf Chowdhury](#)

- Engineered a **dependency-aware execution framework** for LLM-driven knowledge graph **updates**, mapping semantic dependencies into a **Directed Acyclic Graph (DAG)** to isolate conflicts and enable safe parallel batching.
- Deployed a **concurrent reasoning engine** using **Qwen-2.5-7B** and **vLLM** on **NVIDIA A40 GPUs**, optimizing inference throughput while preserving the **sequential semantics required for dependent memory updates**.
- Achieved up to **23% improvement in final-state correctness** under high-contention workloads versus naive batching, **maintaining 100% sequential correctness** while keeping throughput within **3% of unconstrained parallel baselines**.

Adaptive-MAPPO: Multi-Agent Reinforcement Learning Advisor: [Dr. Lei Ying](#)

- Architected **Adaptive-MAPPO**, a **Multi-Agent Reinforcement Learning (MARL)** framework that dynamically interpolates between centralized and decentralized critics, **reducing late-training performance degradation compared to fixed-weight baselines**.
- Engineered a **fully differentiable neural gating mechanism** to calculate per-agent mixing coefficients from normalized value-level features, dynamically balancing the **bias-variance tradeoff** in Centralized Training with Decentralized Execution (CTDE).
- Integrated custom MARL algorithms into **Meta AI's BenchMARL ecosystem**, orchestrating scalable training pipelines across VMAS and PettingZoo environments using **TorchRL, Hydra, and NVIDIA A40 GPUs**.

High Performance CNN Layers in CUDA C++ Advisor: [Dr. Reetuparna Das](#)

- Engineered a **high-performance CUDA C++ implementation for CNN layers**, achieving a **964x speedup** over the naive baseline with a total runtime of **0.014s**; ranked **2nd in class** for execution speed.
- Maximized **V100 microarchitecture throughput** using **FP16 vectorization**, weight-packing memory-layout transformations, and aggressive thread coarsening to saturate floating-point and memory pipelines.

Ocular Disease Diagnosis via CNN & Vision Transformer (ViT)

- Developed a multi-class diagnostic system using **EfficientNetB2** and **Vision Transformer (ViT)** with differential learning rates and a custom PyTorch wrapper (**ash-lib**).
- Achieved a peak **94.82% accuracy and 94.42% macro F1-score** with the ViT model, outperforming the CNN across all metrics on a held-out study dataset.
- Implemented an end-to-end MLOps pipeline with **W&B** for experiment tracking and a **Gradio** deployment featuring a human-in-the-loop data feedback loop.

MUSE: Real-Time AI Audio Personalization Advisor: [Dr. Vera Liao](#)

- Engineered a client-side **TensorFlow.js** personalization engine (MLP) that learns user gain preferences in real-time by mapping 23 extracted spectral features to adaptive audio mixes.
- Designed a **privacy-first architecture** that persists model weights in local storage, solving the “Cold Start” problem for new users via genre-based initialization vectors.
- Optimized system responsiveness based on iterative user feedback, implementing debouncing in the **JavaScript** pipeline to ensure immediate auditory feedback during playback.